1 Introduction

Suppose we want to find the area of the semicircle in the upper half-plane described by the equation $x^2+y^2=a^2$. We can solve for y to realize the semicircle as the graph of the function $y=\sqrt{a^2-x^2}$, and the area is then given by the integral

$$\text{Area} = \int_{-a}^{a} \sqrt{a^2-x^2}dx.$$

If this were $\int_{-a}^{a} x\sqrt{a^2-x^2}dx$, we could evaluate the integral by the substitution $u=a^2-x^2$. However, this will not work here. We now outline a different strategy that works for this integral as well as many similar integrals.

To simplify things, let's consider the indefinite integral

$$\int \sqrt{a^2-x^2}dx$$

with $a>0$.

1. Make the substitution $x=a\sin(\theta)$ and use the Pythagorean identity $1-\sin^2(\theta)=\cos(\theta)$ to simplify the integrand. Assume that $-\frac{\pi}{2}\leq\theta\leq\frac{\pi}{2}$ so that $\sin(\theta)$ is invertible.

2. Now calculate dx in terms of $d\theta$.

3. Use this substitution to rewrite the integral. Evaluate the integral in terms of θ.

4. We have evaluated the integral in terms of θ, but we really want to evaluate it in terms of our original variable x. To help you do so, draw a right triangle (you may assume $0 < \theta < \frac{\pi}{2}$), labeling each side in terms of a and x. Use the triangle to compute θ, $\sin(\theta)$, and $\cos(\theta)$.

5. Substitute your expressions for θ, $\sin(\theta)$, and $\cos(\theta)$ into your answer to Question 3 to evaluate the integral in terms of x.

6. Use your answer to Question 5 to calculate the area $\int_{-a}^{a} \sqrt{a^2 - x^2}dx$ of the semicircle from the previous page. Verify that your answer agrees with the formula you learned in elementary geometry.

2 Trig Substitution

The technique used on the previous page is called *Trig Substitution*. Note that it is actually an example of an *inverse substitution*, since we set $x = g(t)$ for some invertible function g, and used the substitution rule

$$\int f(x)dx = \int f(g(t))g'(t)dt$$

in reverse. We can use the Pythagorean identity $1+\tan^2(\theta) = \sec^2(\theta)$ to make other trig substitutions. These are recorded in the table below:

Table of Trigonometric Substitutions

Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a\sin\theta, \quad -\frac{\pi}{2} \leqslant \theta \leqslant \frac{\pi}{2}$	$1 - \sin^2\theta = \cos^2\theta$
$\sqrt{a^2 + x^2}$	$x = a\tan\theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$1 + \tan^2\theta = \sec^2\theta$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta, \quad 0 \leqslant \theta < \frac{\pi}{2}$ or $\pi \leqslant \theta < \frac{3\pi}{2}$	$\sec^2\theta - 1 = \tan^2\theta$

Note that these also work for expressions such as $\sqrt{a^2 - b^2x^2}$ for $b > 0$, since we can rewrite

$$\sqrt{a^2 - b^2x^2} = \sqrt{b^2\left(\frac{a^2}{b^2} - x^2\right)} = b\sqrt{\frac{a^2}{b^2} - x^2}.$$

3 Practice

7. Use trig substitution to evaluate each of the following integrals.

   (a) $\displaystyle\int \frac{\sqrt{9-x^2}}{x^2}dx$

(b) $\int \frac{1}{x^2\sqrt{x^2+4}}dx$

(c) $\int \frac{dx}{\sqrt{x^2 - a^2}}$, where $a > 0$.